

# Lucky Verma

MACHINE LEARNING ENGINEER | AGENTIC SYSTEMS | KNOWLEDGE GRAPHS | RAG

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## SUMMARY

Machine Learning Engineer with 4+ years building production agentic AI systems, knowledge graphs, and deep learning pipelines for large-scale data infrastructure. Expert in multi-agent orchestration (LangGraph, DSPy), RAG pipelines, entity resolution, model evaluation, and real-time inference serving on GPU clusters.

## TECHNICAL SKILLS

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| AI / LLM  | RAG, Agentic AI, <b>LangGraph</b> , <b>DSPy</b> , LangChain, vLLM, Fine-tuning (LoRA, RLHF/DPO), Tool Use, HF Transformers                                 |
| ML / DL   | <b>PyTorch</b> , TensorFlow, FSDP, DeepSpeed, SLURM Multi-GPU, Triton Kernels, <b>Deep Learning</b> , <b>Computer Vision</b> , NLP, <b>CUDA</b>            |
| Infra     | <b>Knowledge Graphs</b> (Apache AGE, FalkorDB), PostgreSQL, LanceDB, <b>FastAPI</b> , <b>Docker</b> , K8s, Terraform, <b>AWS</b> , Redis, Model Monitoring |
| Languages | <b>Python</b> , SQL, C++, JavaScript, Bash                                                                                                                 |

## WORK EXPERIENCE

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| <b>Eccalon LLC (Defense AI contractor, \$162M+ federal contracts)</b><br><b>MACHINE LEARNING ENGINEER</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <i>Nashville, TN</i><br><i>Jun 2022 – Present</i>           |
| <ul style="list-style-type: none"><li>Architected and deployed <b>AEGIS</b>, a production <b>deep learning</b> + agentic AI system for automated compliance assessment, compressing a <b>40-week manual process</b> to <b>under 24 hours</b> through <b>LangGraph</b> orchestration, <b>DSPy</b> optimization, and 11-stage multi-hop reasoning — serving <b>production deployment</b> across federal data sources (SAM.gov, SEC EDGAR, GLEIF).</li><li>Engineered <b>knowledge graph</b> (PostgreSQL + Apache AGE) with entity resolution across SAM.gov, SEC EDGAR, and GLEIF corporate registries, achieving <b>95%</b> accuracy in multi-hop ownership-chain tracing with sub-10-second query response.</li><li>Built production <b>document intelligence pipeline</b> with GPU-accelerated embeddings (<b>Ollama</b> + NIM fallback), <b>LanceDB</b> vector search, and <b>FalkorDB</b> knowledge graph — reducing multi-source research synthesis from days of manual review to minutes across federal databases and regulatory documents.</li><li>Built <b>model evaluation</b> framework benchmarking 10 LLM baselines across 52 real-world entities with <b>feature engineering</b>, contamination testing, and multi-annotator validation — establishing the first public evaluation methodology for automated compliance systems under <b>32 CFR 117.11</b>.</li><li>Deployed production <b>RAG pipelines</b> on AWS with multi-path reasoning and hybrid vector/keyword search, improving retrieval relevance <b>40%+</b> over keyword-only baseline; built GPU-accelerated CV pipeline (<b>PyTorch</b>, <b>CUDA</b>) with <b>95%+</b> accuracy.</li><li>Owned end-to-end <b>ML infrastructure</b> — Docker multi-service orchestration, Terraform IaC, <b>CI/CD</b>, <b>model serving</b>, <b>monitoring</b>, and multi-backend embedding fallback — maintaining <b>99%+</b> <b>uptime</b> across <b>6 production ML services</b> with <b>distributed systems</b> architecture.</li></ul> |                                                             |
| <b>University of Maryland, Baltimore County</b><br><b>GRADUATE RESEARCH ASSISTANT</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <i>Baltimore, MD</i><br><i>Jan 2022 – May 2023</i>          |
| <ul style="list-style-type: none"><li>Developed ensemble <b>1D U-Net</b> for biomedical signal classification (PPG → sleep staging), achieving <b>92%</b> accuracy; built web-based data science learning platform using CV/ML pipelines with <b>pandas</b> and <b>Streamlit</b>.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                             |
| <b>Vast Dream Group</b><br><b>AI SPECIALIST</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <i>Mumbai, India (Remote)</i><br><i>Jan 2021 – Aug 2021</i> |
| <ul style="list-style-type: none"><li>Built recommendation engine using <b>BART</b> zero-shot classification (96% accuracy on MNLI); architected AI measurement systems for optics industry, contributing to <b>\$1M+</b> revenue pipeline across 3 product lines.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                             |
| <b>TandM Techlabs</b><br><b>SOFTWARE DEVELOPER</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <i>Secunderabad, India</i><br><i>Jun 2018 – Dec 2020</i>    |
| <ul style="list-style-type: none"><li>Built urban mobility platform with geospatial ML: <b>Pandana</b> accessibility metrics, <b>Valhalla/OSMnx</b> route optimization on <b>PostGIS</b>, cached with <b>Redis</b> and containerized on <b>Kubernetes</b> — serving real-time discovery for B2B logistics.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                             |

## KEY PROJECTS & OPEN SOURCE

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| <b>FOCI-Bench: First Benchmark for Foreign Ownership Risk Detection</b><br><i>52 real-world companies   10 baselines   Multi-source entity resolution</i>                                                                                                                                     |               |
| <ul style="list-style-type: none"><li>Created the first public benchmark for evaluating automated FOCI (Foreign Ownership, Control, or Influence) detection systems, covering ownership tracing, beneficial ownership, and state-ownership factors under <b>32 CFR 117.11</b>.</li></ul>      |               |
| <b>InfiniteContext-1B: Million-Token LLM Architecture</b><br><i>DeepSeek-V3 MLA   Distributed FSDP   Consumer GPU inference</i>                                                                                                                                                               | <i>GitHub</i> |
| <ul style="list-style-type: none"><li>Designing MLA-based LLM architecture targeting <b>93%+</b> KV cache reduction for million-token inference on consumer GPUs; implementing custom Triton kernels and FSDP distributed training pipeline on SLURM clusters. <i>[In progress]</i></li></ul> |               |

## EDUCATION

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| <b>University of Maryland, Baltimore County</b><br><i>Master of Science in Computer Science – GPA: 3.97</i>                                                                     | <i>Baltimore, MD</i><br><i>May 2023</i>  |
| <ul style="list-style-type: none"><li><b>DARPA Catalyst Award</b> (\$50K) – Caselet, selected from <b>250+ teams</b> for AI-powered national security workforce tools</li></ul> |                                          |
| <b>SRM University</b><br><i>Bachelor of Technology in Electrical &amp; Electronics Engineering – GPA: 3.9</i>                                                                   | <i>Chennai, India</i><br><i>May 2019</i> |